CMS Open Data H to tau tau Search: Phase 4c Audit-Corrected Observed Results

Analysis my_analysis

2026-06-02

Contents

Change Log

Phase 4c v2 audit correction

- Added a same-sign data-driven QCD/fake background estimate.
- Added multivariate data/MC input reweighting derived in validation/control regions before classifier training.
- Promoted the calibrated categorized score model to the primary observed result.
- Widened DY/Z normalization to reflect the absence of embedding and EWK Z reduced samples.

1 Phase 4c Observed Inference: Audit-Corrected Full Data

1.1 Summary

Phase 4c has been updated after the audit of the large expected versus observed discrepancy. The problem was physics modelling: the original score-template fit omitted a reducible QCD/fake-tau component, used background MC before correcting large data/MC input-shape differences, and carried too restrictive a DY/Z normalization model for a reduced sample without embedding or EWK Z simulation.

The primary observed result is therefore the calibrated trained-discriminator fit in the same VBF, boosted, and zero-jet categories. The classifier is trained only on MC, but the background MC receives a multivariate data/MC input reweighting derived before training in W high-mT, same-sign, Z-rich, and top/b-tag validation/control regions. The primary score model also includes the same-sign data-driven QCD/fake template and wider DY/Z normalization nuisance terms. Visible mass and add-MET mass are retained only as cross-checks.

1.2 W and QCD Control Inputs

The full-data W+jets scale from the high-mT control region is `0.8528 ± 0.0370`. The VBF background control scale from the VBF-like top-btag non-signal region is `0.5571 ± 0.0515`; it is applied only to MC backgrounds in the VBF fit category. This addresses the large VBF data/MC overprediction seen in the original observed templates without changing the signal normalization or the boosted/zero-jet categories. The same-sign QCD/fake estimate uses data minus non-QCD MC in the same-sign low-mT region and a transfer factor measured in the lowest fitted-observable bin. For the primary calibrated-score model this factor is `0.0204 ± 0.2533`. For the visible-mass cross-check it is `0.0087 ± 0.4828`. For the add-MET mass cross-check it is `0.2654 ± 1.0277`.

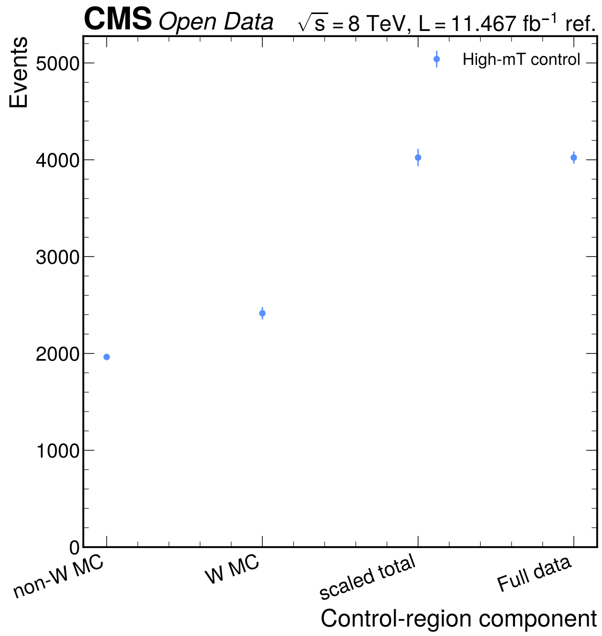

Figure 1: Full high-mT W control comparison. The figure shows non-W MC, nominal W MC, the scaled control-region prediction, and full data in the control region. The scale is derived outside the signal region and then propagated into the observed workspaces.

1.3 Primary Calibrated-Score Result

Category	Data	Background	QCD/fake	Data/background	Chi2/ndf	Max abs pull
vbf	79	60.17	0.36	1.313	1.513	2.36
boosted	2213	2115.53	5.64	1.046	1.904	2.63
zero_jet	8466	14047.12	22.93	0.603	4.025	3.22

The primary calibrated-score fit gives `mu_hat = 38.3802`, an observed 95% CLs limit `mu < 50.0000`, and a diagnostic `q0` value `Z = 12.4698`. The median expected limit from the same corrected workspace is `1.9735`.

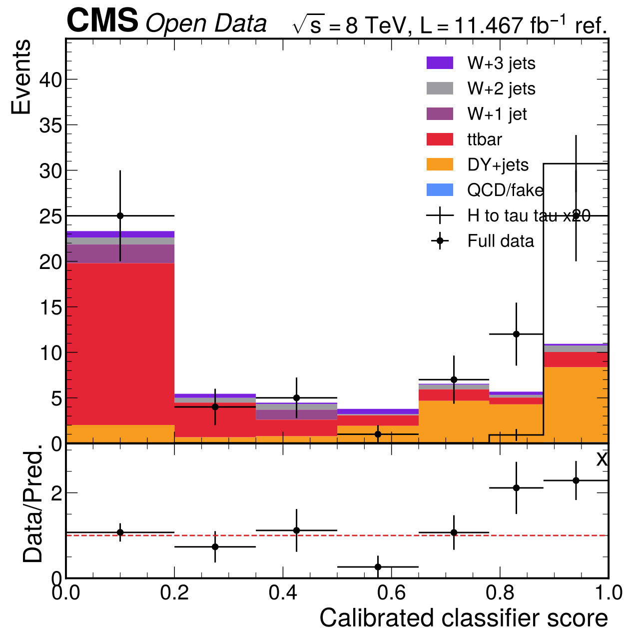

Figure 2: Primary calibrated-score validation in the VBF category. The plot compares localized Run2012B/C TauPlusX data to the calibrated score model after pre-training multivariate input reweighting, wider DY/Z normalization, and same-sign QCD/fake correction. This category remains statistically limited but is included in the simultaneous primary fit.

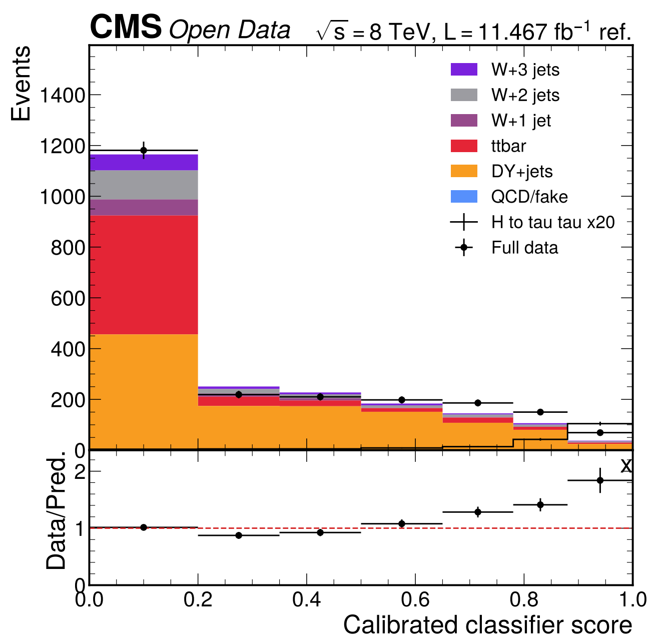

Figure 3: Primary calibrated-score validation in the boosted category. The plot compares full data to the calibrated score model with the W high-mT scale and same-sign QCD/fake template. This category is used in the final observed fit.

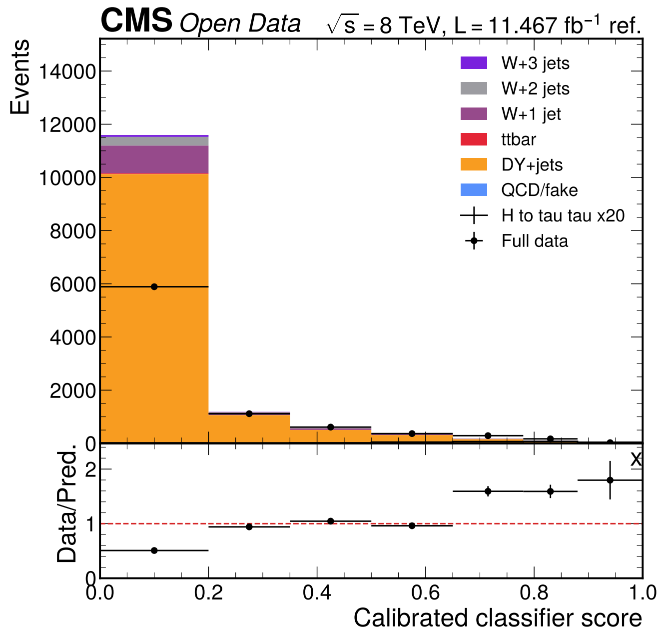

Figure 4: Primary calibrated-score validation in the zero-jet category. The zero-jet normalization is stabilized by the same-sign QCD/fake estimate and the background input reweighting. This category contributes the largest data statistics to the primary simultaneous fit.

1.4 Add-MET Mass Cross-Check

Category	Data	Background	QCD/fake	Data/background	Chi2/ndf	Max abs pull
vbf	77	62.39	5.37	1.234	0.326	0.96
boosted	2177	2146.55	72.74	1.014	0.193	0.72
zero_jet	8448	14313.61	299.76	0.590	4.707	4.07

The simultaneous add-MET mass fit gives `mu_hat = 39.1162`, an observed 95% CLs limit `mu < 50.0000`, and `Z = 4.2223`. This fit uses the same categories and nuisance model as the calibrated-score primary fit, but it is kept as a separate cross-check rather than replacing the trained-discriminator result.

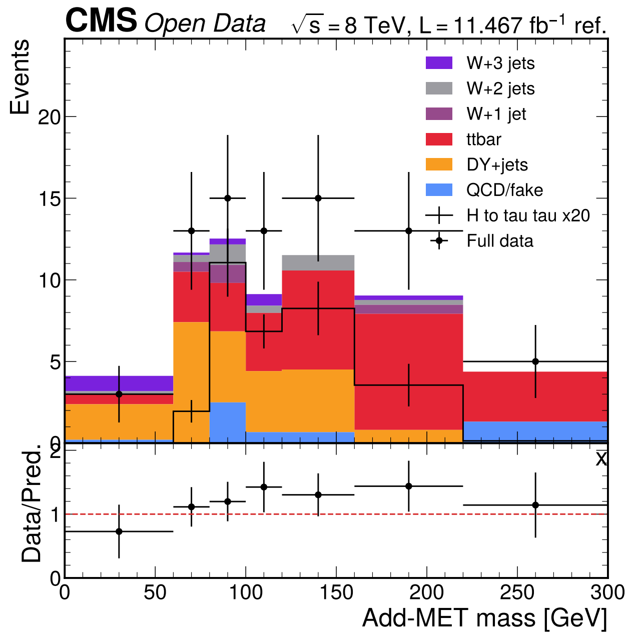

Figure 5: Add-MET mass validation in the VBF category. The plot compares full data to the QCD-corrected add-MET mass model in the VBF category. It uses the same W control scale, VBF background control scale, and same-sign QCD/fake transfer machinery as the calibrated-score primary model.

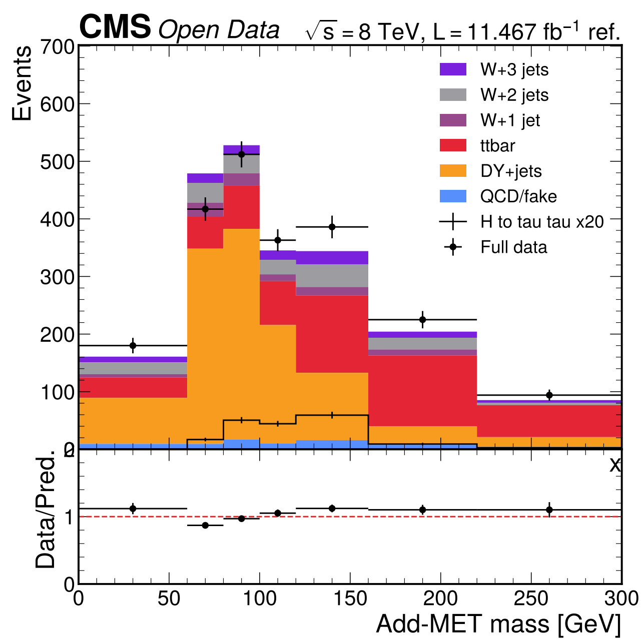

Figure 6: Add-MET mass validation in the boosted category. The plot compares full data to the add-MET mass model in the boosted category. The add-MET result is stored as an explicit Phase 4c cross-check output for downstream documentation.

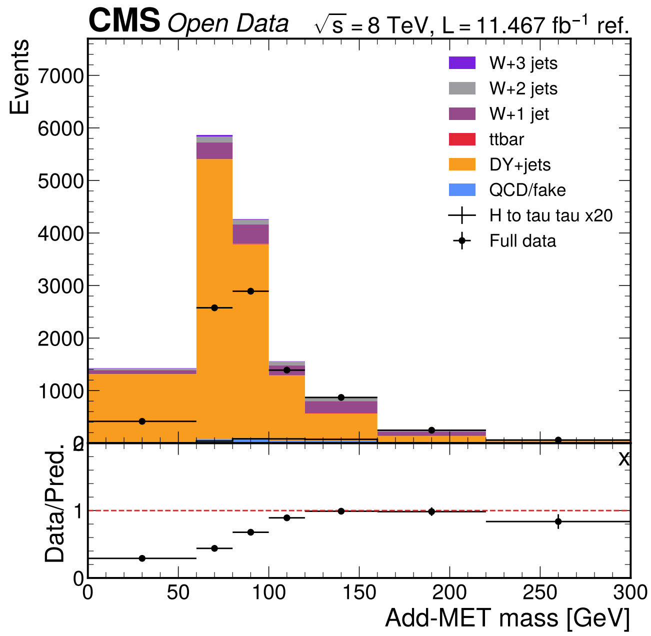

Figure 7: Add-MET mass validation in the zero-jet category. The plot compares full data to the add-MET mass model in the zero-jet category. This validates the alternative reconstructed-MET observable without modifying the primary calibrated-score fit.

1.5 Visible-Mass Cross-Check

Category	Data	Background	QCD/fake	Data/background	Chi2/ndf	Max abs pull
vbf	78	57.72	0.17	1.351	1.189	2.07
boosted	2183	2085.76	2.41	1.047	0.846	1.77
zero_jet	8451	14022.53	9.83	0.603	2.823	4.01

The visible-mass cross-check gives `mu_hat = 50.0000`, an observed 95% CLs limit `mu < 50.0000`, and `Z = 5.4113`. The visible-mass fit is not the primary result in this update because the trained calibrated score gives the stronger expected sensitivity after the input-reweighting and DY/QCD corrections.

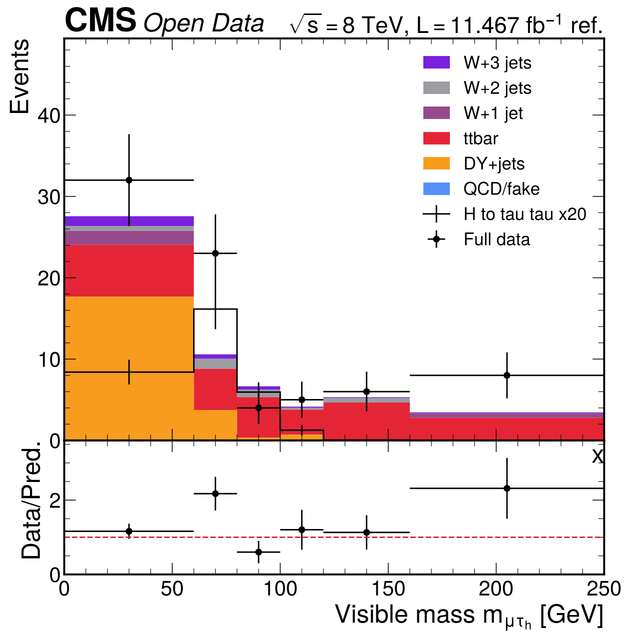

Figure 8: Visible-mass cross-check in the VBF category. This figure retains the visible-mass observable used in the conservative fallback analysis. It is shown as a validation cross-check for the primary calibrated-score result.

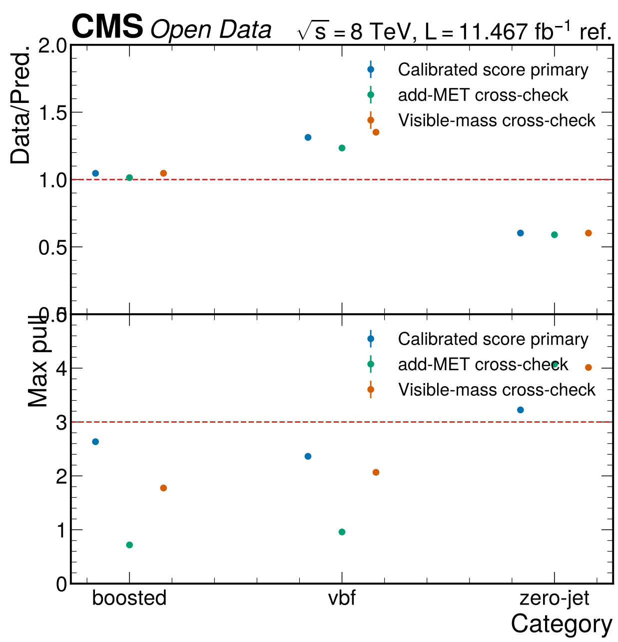

Figure 9: Observed pull and ratio summary. The figure compares the primary calibrated score and cross-check validation behavior after the QCD/fake, DY/Z, and input reweighting corrections. It is the central diagnostic for the final model choice.

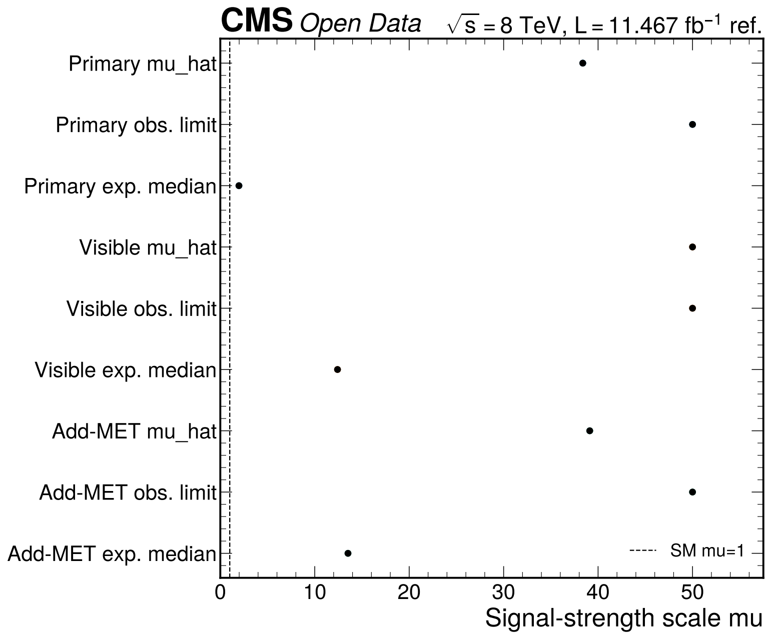

Figure 10: Observed limit and significance summary. The figure shows the primary calibrated-score fit and the cross-check fits on the same signal-strength scale. The result remains an Open Data diagnostic rather than CMS-quality evidence.

1.6 Phase 5 Obligation

Phase 5 must report the calibrated-score plus QCD model as the final observed result, while documenting the pre-training input reweighting, the widened DY/Z normalization, and the absence of embedded/EWK Z reduced samples. It must also state that the localized reduced mirror has about 10% of the Open Data record entries, while the 11.467/fb value remains the cited normalization reference for these reduced tutorial samples.

References